Alice V. Pierce

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EDUCATION

University of California, Davis Expected March 2027
Ph.D. in Plant Biology · Designated Emphasis in Biotechnology
Advisor: J. Grey Monroe, Department of Plant Sciences
Dissertation: Associations between intron architecture, genome evolution and gene regulation

University of California, Davis June 2019
B.S. in Biotechnology (Honors) · Minor in Computational Biology
Advisor: Alan Rose, Department of Molecular and Cellular Biology
Honors thesis: Understanding what kind of sequences can support transcription initiation in response to stimulatory introns

SUMMARY

Computational biologist investigating how noncoding sequences and gene architecture are associated with chromatin, gene expression and genome evolution. Experienced in integrating multi-omics data, applying statistical and machine learning models, and building reproducible workflows for comparative genomics. Committed to advancing reproducible and inclusive science through research, mentorship, and outreach.

SKILLS

Computing	R, Python, Bash, SLURM/HPC, Snakemake, Git/GitHub
Bioinformatics	ChIP-Seq, ATAC-Seq, RNA-Seq analysis, alternative splicing analysis, gene annotation, comparative genomics, phylogenomics, machine learning
Languages	English, Spanish

RESEARCH EXPERIENCE

Graduate Student Researcher Grey Monroe Lab, Dpt. Plant Sciences Jan 2022 – Present UC Davis, CA

Dissertation Research:

- **Associations between intron architecture, chromatin and expression features in *Arabidopsis thaliana***
  - Integrated ChIP-seq, ATAC-seq and RNA-seq data using custom R and Python scripts. Applied regression and random forest models to model associations between intron features, active chromatin marks and expression breadth.
- **Evolution of spliceosomal intron architecture and sequence across eukaryotic genomes**
  - Built scalable Snakemake workflows with Bash and Python to extract intron features from more than 5,000 eukaryotic genomes. Used comparative and phylogenetic analyses to examine variation in intron architecture and k-mer profiles across lineages to gain evolutionary insights into intron conservation and genome complexity in eukaryotes.
- **Persistence and regulatory potential of spliceosomal introns**
  - Developing a review synthesizing evidence on the evolutionary cost of introns, the processes that may explain their persistence, and how intron sequence and organization interact with chromatin, transcription, and splicing.

Other Projects:

- Association between histone post-translational modifications and double-strand break repair pathway choice using CRISPR-Cas9
- Evolution of epigenome-associated DNA repair proteins (MSH6, PDS5C)

Translational Data Science Intern Calico Life Sciences June – Sep 2025 South San Francisco, CA

- Built a differential splicing module using Bash, Python, and R to enhance internal RNA-seq analysis workflows for cancer genomics data, improving the detection of alternative splicing diversity
- Validated results against external datasets to strengthen exon-level quantification benchmarks, enhancing accuracy and interpretability of splicing event detection within translational data workflows

Junior Specialist Maeli Melotto Lab, Dpt. Plant Sciences Oct 2019 – June 2021 UC Davis, CA

- Created an effective protocol for isolating mesophyll protoplasts from *Arabidopsis thaliana* and *Lactuca sativa.*
- Developed high-throughput GUS staining methods to observe the spatiotemporal expression of the JAZ4 protein throughout the Arabidopsis life cycle
- Performed yeast two-hybrid to screen for hormone-induced JAZ4-protein interactions

Undergraduate Student Researcher Alan Rose Lab, Dpt. Mol. and Cel. Biology Jan 2018 – Aug 2019 UC Davis, CA

- Leveraged recombinant DNA technologies to investigate sequences that support transcription initiation in response to stimulatory introns

PUBLICATIONS

2025 **Pierce A,** Rose A, Monroe G. "Intron architecture predicts chromatin features in Arabidopsis thaliana." In Revision at Plant Physiology; bioRxiv. https://doi.org/10.1101/2025.10.15.682614

2025 Quiroz D, **Pierce A**, Monroe G. "Evolutionary trajectories of cohesin gatekeepers across the Tree of Life." bioRxiv. https://doi.org/10.1101/2025.05.08.652912

2025 Monroe JG, Lee C, …, **Pierce A**, et al. "Convergent Evolution of Epigenome Recruited DNA Repair across the Tree of Life." eLife. https://doi.org/10.7554/eLife.105016.1

2024 Quiroz D, Lopez-Mateos D, Zhao K, **Pierce A**, et al. "H3K4me1 recruits DNA repair in plants." The Plant Cell. doi:10.1093/plcell/koae089

2024 **Pierce A**, Anderson I, Sinha N, Monroe G. "Celebrating modern plant diversity cataloging." Plant Sciences Bulletin

2022 **Pierce A**, Student J, Melotto M. "The Farm-to-Fork Journey: Keeping Produce Fresh and Safe to Eat." Front. Young Minds 10:587135

2021 DeMott L, Oblessuc P, **Pierce A**, et al. "Spatiotemporal regulation of JAZ4 expression and splicing contribute to ethylene- and auxin-mediated responses in Arabidopsis roots." Plant J. 108:1266–1282

IN PREP

Pierce, et al. 2027 “Comparative survey of intron architecture and sequence across the Tree of Life.”

Pierce, et al. 2027 “Why introns persist: the evolution, architecture, and regulatory potential of introns.”

TEACHING EXPERIENCE

Fall 2022, 2023, 2026 Teaching Assistant — Plant Genetics, UC Davis

May 2025 Workshop Facilitator – “How to make a personal website using GitHub”, GSPD SACNAS, UC Davis

Fall 2025 Guest Lecture — "The Role of Introns in Plant Chromatin", Plant Genetics, UC Davis

Winter 2025 Guest Lecture — "Becoming a Latina Scientist", Becoming a Latino Scientist Seminar, UC Davis

June 2024 Workshop Facilitator – “How to make an effective graphical abstract or scientific illustration”, Plant Biology 2024. Over 150 attendees.

Summer 2021 Guest Lecture — "Is the Central Dogma Actually a Dogma?", American Chemical Society

Winter 2017 Learning Assistant — Biodiversity and the Tree of Life, UC Davis

ORAL PRESENTATIONS

Oct 2025	"The Role of Introns in Plant Chromatin." SACNAS 2025, Columbus, Ohio
Sep 2025	"The Role of Introns in Plant Chromatin." International Student Workshop, Nara, Japan (UC Davis–IGDB–NAIST)
March 2025	"The Role of Introns in Plant Chromatin." Annual Biotechnology Program Training Retreat, UC Davis
June 2024	“Exploring the Impact of Intron Architecture on Chromatin Features in Plants”, Plant Biology 2024, Honolulu, Hawaii
Apr 2024	“The Role of Introns in Plant Chromatin.” UC Davis Plant Biology Graduate Group Seminar
Oct 2023	"Understanding the role of introns on the histone code." SACNAS 2023, Portland, Oregon
March 2022	“How does IME affect gene body mutations?” UC Davis Plant Biology Graduate Group Seminar

POSTER PRESENTATIONS

Oct 2025	"The Role of Introns in Plant Chromatin." SACNAS 2025, Columbus, Ohio
Sep 2025	"The Role of Introns in Plant Chromatin." International Student Workshop, Nara, Japan (UC Davis–IGDB–NAIST)
April 2025	“Exploring the Impact of Intron Architecture on Chromatin Features in Plants.” Bay Area Plant Hub Symposium, UC San Francisco
April 2025	“Exploring the Impact of Intron Architecture on Chromatin Features in Plants.” Plant Sciences Symposium at UC Davis
Nov 2024	“Exploring the Impact of Intron Architecture on Chromatin Features in Plants.” SACNAS 2024, Phoenix, AZ
Jun 2024	“Exploring the Impact of Intron Architecture on Chromatin Features in Plants.” Plant Biology 2024, Honolulu, Hawaii
Aug 2023	“Intron Loss and Gain: Implications for Chromatin Biology in Arabidopsis Gene Duplicates.” Plant Biology 2023, Savannah, Georgia
May 2023	“How does intron loss/gain affect chromatin marks in gene duplicates?” Plant Sciences Symposium at UC Davis
March 2023	“How does intron loss/gain affect chromatin marks in gene duplicates?” Annual Biotechnology Program Training Retreat
May 2022	“How do IME introns contribute to the mutation rate and epigenetic landscape of their gene bodies?” Plant Sciences Symposium at UC Davis
April 2019	“What Kind of Sequences Support Transcription Initiation in Response to Stimulatory Introns?” Undergraduate Research Conference at UC Davis

ACADEMIC SERVICE

- Poster Judge for Undergraduate and Master’s student posters in plant sciences and botany, genetics, ecology and evolutionary biology, general biology, and genomics, SACNAS 2026
- Invited Panelist, “Fundraising for Your Chapter: Strategies, Stories, and Sustainable Support”, SACNAS 2026
- Secretary and Organizer, UC Davis Plant Sciences Symposium Organizing Committee, 2022–2024
- Seminar Committee Co-Chair, Plant Biology Graduate Student Association, UC Davis, 2022–2023
- Plantae Fellow, American Society of Plant Biologists (ASPB), 2022–2023

MENTORSHIP

- Undergraduate Research Mentor, Grey Monroe Lab, UC Davis. Trained four undergraduates in molecular biology techniques and guided their independent research and poster presentations.
- Mentor, MESA Undergraduate Research Scholars, June 2023–September 2023
- Mentor, Research e-Mentorship Program, Biotechnology Academy at Sheldon High School and UC Davis Biotechnology Program, Sep 2022–May 2024

LEADERSHIP & OUTREACH

President Graduate Student & Postdoctoral SACNAS Chapter at UC Davis September 2024 – August 2026

- Oversaw activities across seven committees serving 200+ members, strategically guiding professional development, outreach, mentorship, and community-building
- Fostered partnerships with campus organizations, academic departments, and national SACNAS leadership to strengthen cross-disciplinary engagement and resource sharing
- Coordinated outreach events with local middle school, high school, and undergraduate education programs to expand STEM engagement to the local Sacramento Valley community

Program Coordinator & Chapter Lead EnvironMentors at UC Davis Mar 2023 – August 2026

- Directed mentorship programs pairing high school students with graduate student mentors to perform independent research
- Designed and facilitated educational workshops and lesson plans supporting college preparation, scientific thinking, and professional development

Previous roles: President & High School Outreach Co-Chair (2024–2025), Secretary & High School Outreach Co-Chair (2023–2024) at GSPD SACNAS at UC Davis; Mentoring Committee Co-Chair (2023–2024) at Plant Biology Graduate Student Association; Pen Pal, Letters to a Pre-Scientist (2023–2024); Outreach Committee Member, Science Says, UC Davis (2022–2023)

HONORS & AWARDS

July 2026	Root & Shoot Travel Award, ASPB Plant Biology Conference ($4000)
October 2025	Graduate Student Oral Presentation Award, SACNAS 2025
October 2025	Chapter Officer Leadership Retreat Participant Travel Scholarship, SACNAS 2025
September 2025	International Student Workshop, UC Davis – IGDB – NAIST
March 2025	UC Davis College of Biological Sciences DEIJ Initiative Grant, on behalf of GSPD SACNAS Chapter High School Outreach Committee ($500)
March 2025	STEM Talk 2nd Place Award, Annual Biotechnology Graduate Program Training Retreat
October 2024	Chapter Officer Leadership Retreat Participant Travel Scholarship, SACNAS 2024
June 2024	Corteva-ASPB Student Showcase Travel Award, ASPB 2024
October 2023	Travel Scholarship Award, SACNAS 2023
June 2023	Recognition Travel Award, Equity, Diversity, and Inclusion Committee, ASPB 2023
Summer 2023	Summer Graduate Student Research Award for Engineering and Computer Science Applications and Methods ($15,270), UC Davis
2022–2023	General Member Award for Outstanding High School Outreach, GSPD SACNAS at UC Davis
2023	Ford Foundation Fellowship, Honorable Mention, National Academy of Sciences, Engineering, and Medicine
2022	UC Davis Jastro-Shields Research Award ($3,000)
2022–26	UC Davis Department of Plant Sciences Graduate Student Research Award ($80,000)
2021–22	UC Davis Dean's Distinguished Graduate Fellowship ($27,000)
2019	UC Davis Department of Plant Sciences Citation Award
2018	California Crop Improvement Association Scholarship ($3000)
2018	Knowles Agronomy Scholarship ($2500)
2018	Academic Excellence Award, UC Davis Center for Chicanx and Latinx Academic Student Success
2016	William and Charlotte Lider Scholarship